QUICKSTARK.AI · ARCHITECTURE PLAN

WhatsApp Assistant Business Dashboard

How businesses will sign up, set up their WhatsApp AI assistant in one click, and control everything themselves from an online dashboard.

OPTION A

Add it to quickstark.tech

A new "WhatsApp Assistant" section inside the existing QuickStark app.

Recommended

OPTION B

Build a new, separate site

Its own website, login and hosting, sharing the same bot engine.

Possible, slower

What is inside

1. The goal and what every business will control
2. How the system works today
3. Option A: add it to quickstark.tech (sketch and explanation)
4. Option B: a new site (sketch and how we would go about it)
5. Side-by-side comparison and best odds
6. Data, security and the one-click setup flow
7. Build plan and running costs
8. Usage and credits: the cost calculation, pricing and how to build it
9. Risks and decisions for you

1 · THE GOAL

Every business runs its own assistant, without calling us

Today you add and edit businesses yourself through n8n forms. The dashboard moves that control to the business owner: they log in, set up in one click, and manage their bot, products and leads from a phone or laptop.

What each business will control from its dashboard

1. Account and one-click setup

- Sign up and log in (Google or email)
- Setup wizard: business details, products, alert contacts
- Pay, scan the QR code, bot goes live

2. WhatsApp connection

- Live status: connected or disconnected
- Reconnect: show a fresh QR code
- Disconnect or switch number

3. The AI agent

- Switch the bot on or off (pause)
- About, FAQs, tone, tagline, opening hours
- Menu options 1–4, contact email and text line
- Advanced: custom AI instructions

4. Products and inventory

- Add, edit or delete products and services
- Prices, stock, descriptions
- Payment link created automatically per product

5. Leads and activity

- Leads, “person requested”, new contacts
- One-tap “Reply on WhatsApp”
- Mark a lead as handled

6. Reports, alerts and billing

- Daily and weekly numbers, charts
- Alert numbers, alert email, daily report on or off
- Plan, payments, renewal date
- Staff numbers (who can use the stock desk)

> **Key idea:** the bot engine in n8n already reads everything from the database on every message. So any change a business saves in its dashboard is live on WhatsApp on the very next message. No redeploys and no work from you.

2 · HOW IT WORKS TODAY

The current system (already built and live)

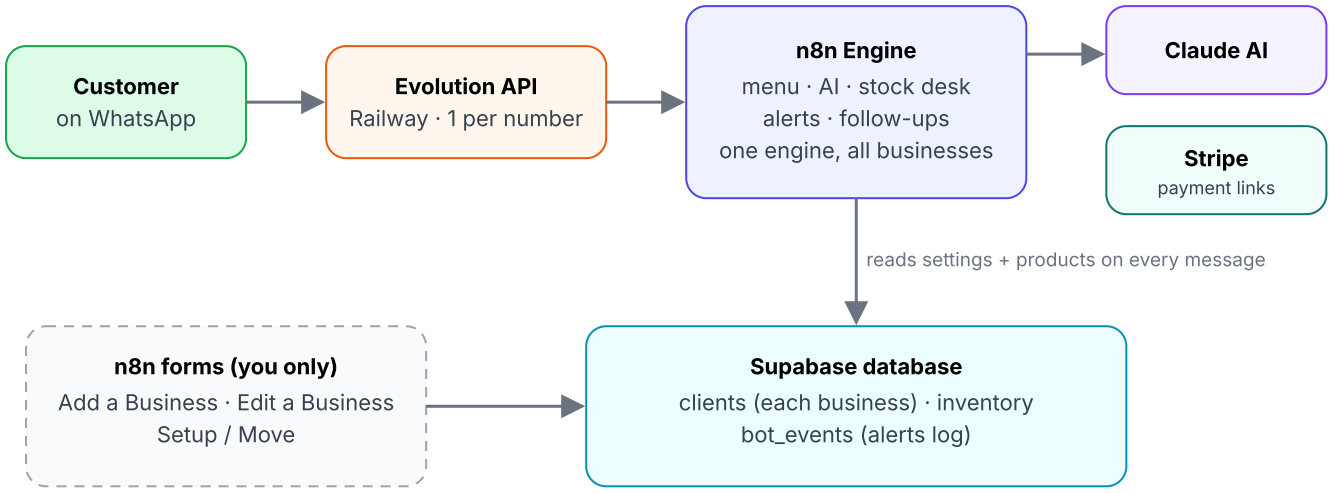


Figure 1 · Today. Only you can add or change a business (through n8n forms).

What we keep

- **The n8n engine stays as it is.** It already serves many businesses and reads everything from Supabase.
- **The Supabase tables stay.** We add a few columns (who owns each business, its status and plan) and security rules.
- **Evolution, Claude and Stripe stay.** The dashboard just calls them on the business's behalf.
- **The n8n forms stay as your admin tools.** You can still add or fix any business yourself.

> The dashboard is a new **front door** to the same system. It does not replace the engine, so the risk to the live bot is low.

3 · OPTION A

Add a “WhatsApp Assistant” section to quickstark.tech Recommended

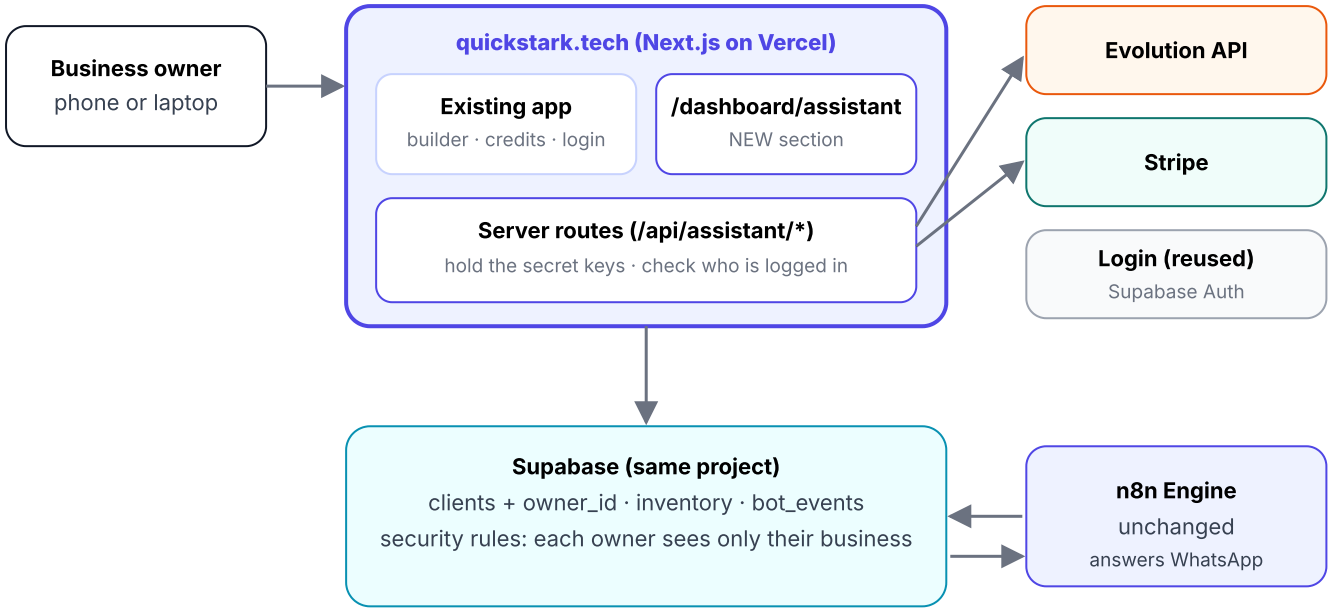


Figure 2 · Option A. One new section inside the app you already run. Everything else is reused.

How it works

- Business owners log in with the **same QuickStark login** (Google, email, phone).
- A new menu item, **WhatsApp Assistant**, opens their dashboard.
- Server routes inside the app talk to Evolution and Stripe. The secret keys stay on the server, never in the browser.
- The dashboard saves to the same Supabase tables the engine reads, so changes go live instantly.
- Stripe payments reuse the app's existing checkout and webhook code.

Why it is the best fit

- **About 70% already exists:** login, hosting, database, Stripe, design system, email.
- **One product to sell:** every bot customer also sees the QuickStark builder, and the other way round.
- **One place to maintain:** one deploy, one login system, one bill.
- **Fastest:** about 2–3 weeks to a working version.

Trade-offs

- The bot product shares QuickStark's brand and domain.
- A bug in the main app could affect the dashboard (and the other way round). We reduce this by keeping the section separate in the code.

4 · OPTION B

Build a new, separate site for the assistant

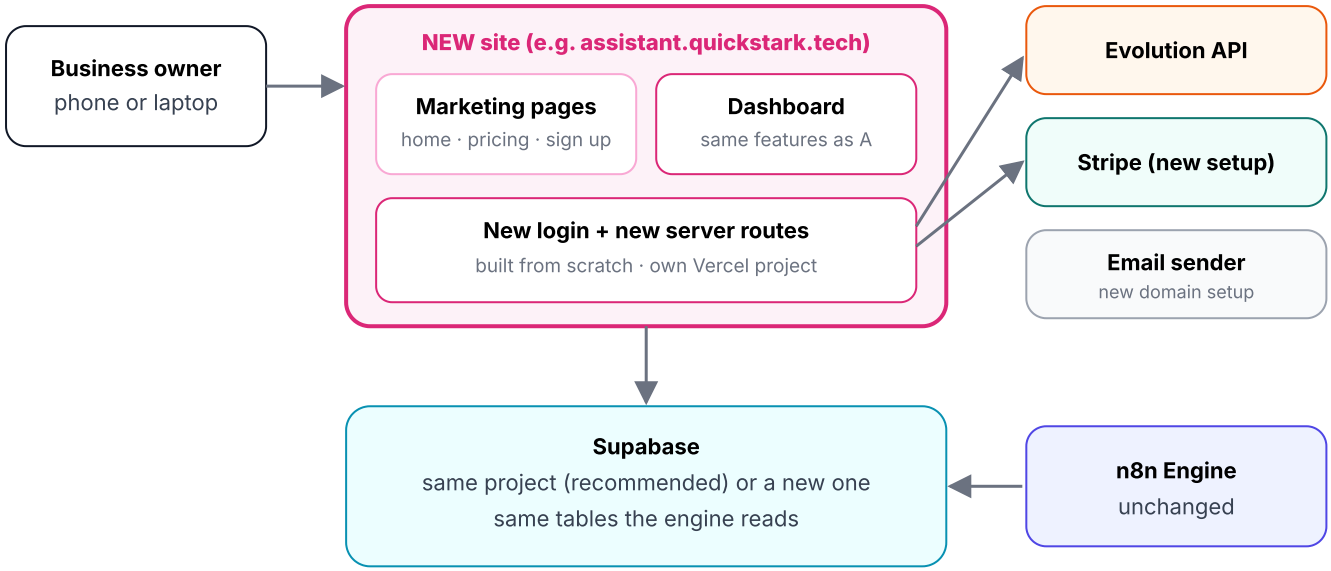


Figure 3 · Option B. A separate site with its own login and hosting, still powered by the same engine.

How we would go about it

1. **Name and domain.** Pick a brand (or use `assistant.quickstark.tech`), set up DNS and SSL.
2. **New project.** A new Next.js app in its own folder or repo, its own Vercel project and environment keys.
3. **Login.** Supabase Auth with Google and email, plus a new sign-up, reset-password and email-verification flow.
4. **Design system.** Copy or rebuild buttons, forms, layout and theme from QuickStark.
5. **Dashboard.** The same screens and server routes as Option A.
6. **Payments.** Stripe checkout, webhook and receipts set up again for the new domain.
7. **Marketing site.** Home, pricing, FAQs, terms, privacy and SEO pages.
8. **Launch.** Email sending for the new domain, monitoring, analytics.

When B makes sense

- You want the bot as its own brand, sold separately from QuickStark
- You plan to sell or spin off the bot business
- A different team will run it

Costs of B

- About 4–6 weeks instead of 2–3
- Two apps, two logins, two deploys to maintain
- Customers of one do not see the other

5 · COMPARISON

Side by side, and the best odds

Factor	A · Add to quickstark.tech	B · New site
Time to a working version	**2–3 weeks**	4–6 weeks
Login, hosting, database	**Reused (already live)**	Rebuilt or re-set-up
Stripe payments	**Reuse existing checkout and webhook**	Set up again
Running cost per month	**No new hosting cost**	Extra Vercel project, domain, email
Maintenance	**One app**	Two apps
Cross-selling	**Bot buyers also see the builder**	Separate audiences
Own brand and identity	Shares QuickStark brand	**Fully separate brand**
Selling it separately later	Possible, needs splitting out	**Easy**
Risk to the live bot	**Low (engine unchanged)**	**Low (engine unchanged)**
Risk of delays	**Low**	Medium

OPTION A

85%

Estimated chance of shipping a solid version on time. Most parts are already built and tested in production.

Best odds · Recommended

OPTION B

60%

More new parts (login, payments, domain, marketing site) mean more places for delays and bugs.

Choose only for a separate brand

Recommendation: build **Option A** now. Keep the dashboard code in its own folder (`src/app/dashboard/assistant` and `src/lib/assistant`). If you later want a separate brand, that folder can move to a new site with little rework, so A does not close the door on B.

The odds are planning estimates based on how much already exists, not guarantees.

6 · DATA AND SECURITY

What changes in the database

Table	Change	Why
`clients`	Add `owner_id` (the logged-in user), `status` (setup, awaiting payment, live, paused), `plan`, `paid_until`, `whatsapp_status`	Know who owns each business and whether the bot should answer
`inventory`	No new columns	Already holds prices, stock and payment links
`bot_events`	Add `handled` (yes/no)	Let owners tick off leads in the dashboard
`business_members` (new)	Extra people allowed into a business (staff, manager)	Teams, not just one owner
`assistant_payments` (new)	Each Stripe payment for a bot plan	Billing history and renewals

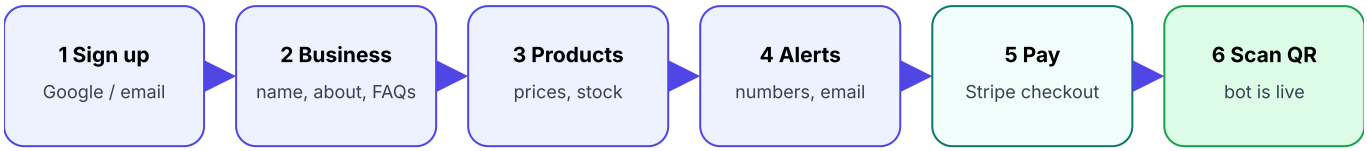
Security rules (the most important part)

- **Each owner sees only their own business.** Supabase row-level security on every table: `owner_id = the logged-in user` (or a listed member).
- **Secret keys never reach the browser.** The Evolution key, Stripe secret and service keys live only on the server (Vercel environment).
- **The engine keeps full access.** n8n uses its own server login, so the new rules do not affect the live bot.
- **Sensitive fields are server-only.** A business cannot change its own `status`, `plan` or `paid_until`; only the payment webhook can.
- **Limits.** One WhatsApp number per business on the starter plans, rate limits on QR requests and saves.
- **Paused or unpaid bots go quiet.** The engine already skips businesses that are not active; we connect that to `status`.

> **Before any change:** we test the security rules on a copy of the database (a Supabase branch) so the live bot is never at risk.

6 · KEY FLOWS

One-click setup, step by step



Steps 2–4 are pre-filled with smart defaults, so a business can finish in about 3 minutes.

What happens behind the “one click”

1. The dashboard saves the business and its products to Supabase (status: *awaiting payment*).
2. Stripe checkout opens for the chosen package. After payment, the webhook marks it *paid* and records the payment.
3. The server creates the business's Evolution instance and points it at the n8n engine (the same steps the Add a Business form does today).
4. The dashboard shows the QR code and refreshes it until WhatsApp connects. Then status becomes *live*.
5. Each product with a price gets a Stripe payment link automatically.
6. The owner gets a “You're live” WhatsApp message and email with a short getting-started guide.

Everyday control (all instant)

Owner does	What happens
Changes a price	Saved to `inventory`; the bot quotes the new price on the next message; payment link updated
Pauses the bot	Status set to *paused*; the engine stops replying; the owner can reply personally
Edits FAQs or tone	Saved to `clients`; the AI uses it on the next message
WhatsApp disconnects	Dashboard shows a red banner and a “Reconnect” button with a new QR code; owner also gets an alert
Ticks a lead as handled	It leaves the “To follow up” list and the daily report

6 · SCREENS

Dashboard sketch

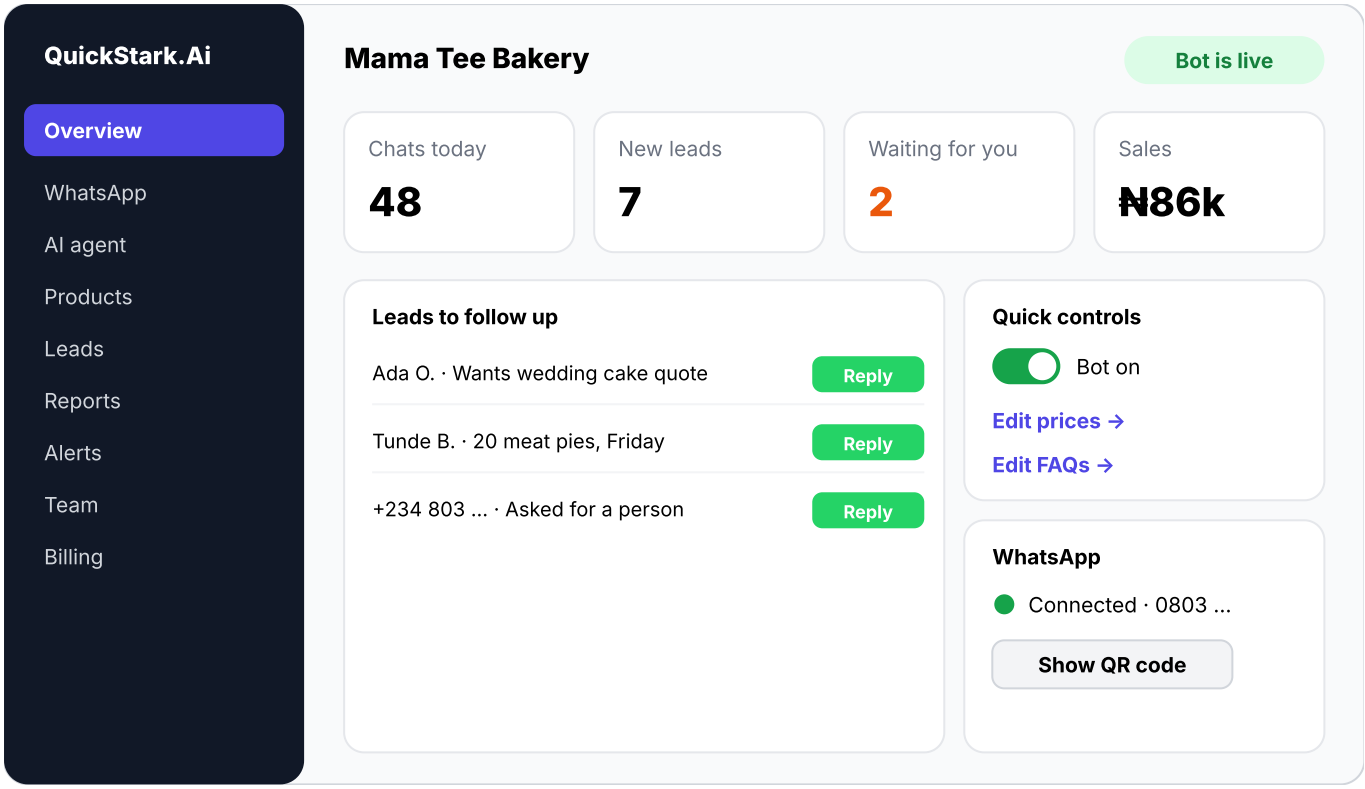


Figure 4 · Overview screen (sketch). Works on phones: the side menu folds into a bottom bar.

Products
Table with price, stock and payment link; add with one form; bulk upload from a spreadsheet later.

AI agent
About, FAQs, tone, menu, hours, contact email and text line, with a live “preview reply” box.

Billing
Plan, next renewal, receipts, upgrade (for example Starter to Business website).

7 · BUILD PLAN

How we would build Option A

Phase	What	Done when	Time
1	Foundation: database columns, security rules (tested on a copy), new dashboard section and menu item	A logged-in user sees only their own business	2–3 days
2	One-click setup: wizard, Stripe checkout, Evolution instance, live QR code	A new business can go from sign-up to live bot alone	4–5 days
3	Control: AI agent editor, products and inventory, pause and resume, contacts and alerts	Every setting in today's Edit form is in the dashboard	3–4 days
4	Leads and reports: overview numbers, leads list with Reply and Handled, charts	Owners can work their leads from the dashboard	2–3 days
5	Billing and renewals: payments history, renewal reminders, pause when unpaid	Payments run without you	2 days
6	Polish and launch: phone layout, emails, help pages, test with 2–3 real businesses	First paying businesses onboarded	2–3 days

Total: about 2–3 weeks of build time. Each phase ships on its own and can be tested before the next.

Running costs (monthly, rough)

Item	Option A	Option B
Hosting (Vercel)	Already paid	+1 project (often free to $20)
Database (Supabase)	Already paid; grows slowly	Same, or +$25 for a new project
n8n executions	1 per WhatsApp message (plan limit to watch)	Same
Evolution on Railway	Grows with numbers connected	Same
Claude AI	Small cost per AI answer	Same
Domain and email	None new	New domain and email sender

8 · USAGE AND CREDITS

The calculation: what one AI reply really costs

Built from **your bot's real runs** (token counts read from the n8n executions on 21 Oct 2026) and current prices. Estimates; re-run after 2–4 weeks of metering.

Inputs	
AI model today	Claude Sonnet 5.5 · $2 in / $10 out per million tokens
Cheap model	Claude Haiku 5.5 · $0.10 in / $0.50 out
Cache read	$0.20 per million (Sonnet 5.5)
Measured per AI call	5,700–6,200 tokens in · 100–400 out (system prompt is ~95%)
Services lookup	2 AI calls (tool), ≈ double cost

Inputs	
Voice notes	OpenAI transcription ≈ $0.006/min
n8n Cloud Pro	≈ €60/month for 10,000 executions → ≈ ₦9.8 per execution
Executions	≈ 2 per customer message today (incoming + the bot's own echo)
Exchange rate	₦1,390/$ today (parallel); **₦1,738/$** for pricing (+25% buffer)
Message mix	assumed, see shares below

Cost per AI reply, by type (AI only, at ₦1,390/$)

Reply type	Share	Credits	Today	Optimised
Simple question	65%	1	₦24	**₦0.5**
Services/price lookup (tool: 2 AI calls)	15%	1	₦51	**₦13**
Photo the AI looks at	8%	2	₦31	**₦16**
Voice note (transcribe + answer)	7%	2	₦29	**₦15**
Long chat + lookup (worst case)	5%	2	₦63	**₦21**

Optimised = prompt caching on the business instructions + the services list put straight into the prompt (no second AI call) + Claude Haiku 5.5 for simple questions, Sonnet 5.5 for the rest. Average today **₦31** per reply → optimised **₦5.7** (about 80% less).

8 · USAGE AND CREDITS

Where the money goes, and how not to cheat yourself

Setup	AI / reply	n8n / reply	Total / reply (avg)	Expensive replies	1,000 replies / month
Today	₦31	₦27	₦58	₦78	**₦59,157**
AI optimised, n8n Cloud	₦5.7	₦27	₦33	₦44	**₦33,668**
AI optimised + self-hosted n8n	₦5.7	₦0 (flat fee)	₦5.7	₦16	**₦7,736**

1,000 replies also brings about 1,400 menu and option messages (no AI). Fixed costs (Evolution, self-hosted n8n ≈ $30/month) are shared across 20 businesses.

Finding 1 · n8n is the hidden cost. On n8n Cloud every message costs ≈ ₦20 in executions, more than the AI after optimisation. Fix: **self-host n8n** (flat ≈ $20/month, unlimited runs) and stop the bot's own messages from triggering a run.

Finding 2 · one-off fees don't cover running costs. A ₦100,000 one-time setup with free AI forever pays for a business doing 1,000 replies a month for only **1.7 months** today, or **13 months** on the best setup. Keep the setup fee, and add a **monthly AI plan**.

Recommended monthly AI plans (best setup, priced at ₦1,738/$)

Plan	Includes	Price	Your cost (incl. fees)	Margin
Lite	500 AI replies	**₦20,000**/month	₦6,940	**65%**
Standard	1,500 AI replies	**₦45,000**/month	₦15,006	**67%**
Busy	4,000 AI replies	**₦100,000**/month	₦34,870	**65%**

Top-up packs (never expire)		
+250 replies	**₦15,000**	₦60 per reply
+1,000 replies	**₦55,000**	₦55 per reply
+3,000 replies	**₦150,000**	₦50 per reply

The safety rule: price every reply at ≥ 3× the cost of an *expensive* reply (≈ ₦20 at the pricing rate → ≥ ₦61), not the average. Plans keep a 65%+ margin. Heavy replies (photo, voice, long chats) use 2 credits. Menu, prices list and option 5 are free.

8 · USAGE AND CREDITS

How credits work, end to end

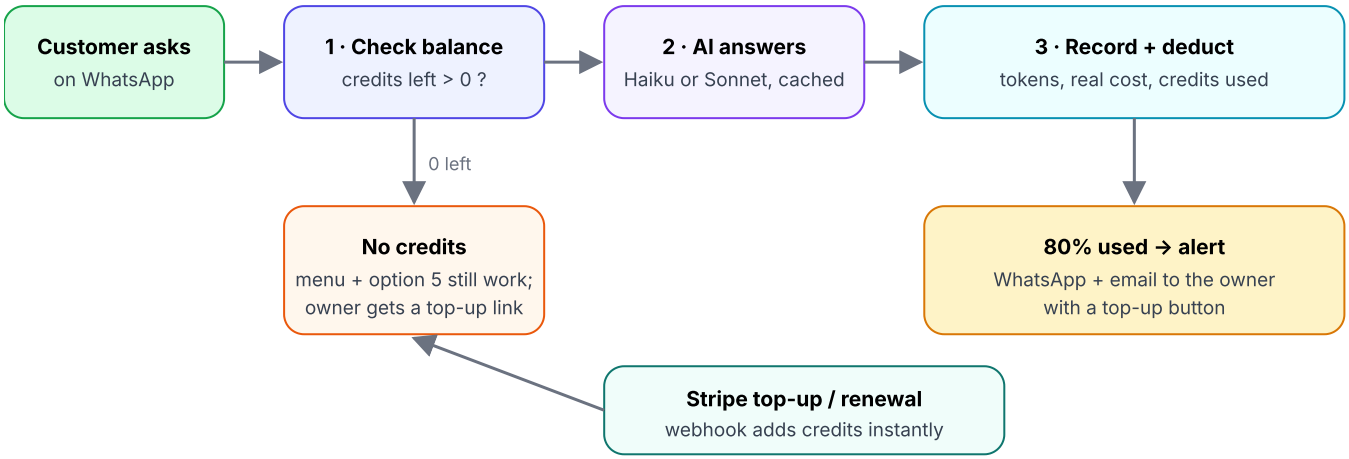


Figure 5 · One balance per business. The bot never goes silent; only AI answers pause at zero.

What the owner sees in the dashboard

- **Usage bar:** “640 of 1,500 replies used · resets 1 Nov”, plus a daily chart and a text / photo / voice breakdown.
- **Top up** button, **auto top-up** switch (“add 1,000 when below 50”), receipts.

What you see in your admin view

- Per business: replies, real AI cost, n8n runs, revenue, **margin**, updated daily.
- Red flag when a business's cost per reply or margin crosses a limit, so you can move it to a bigger plan (fair-use clause in your terms).
- Monthly check of the exchange rate and AI prices; top-up prices can be adjusted for new purchases.

8 · USAGE AND CREDITS

How we would actually implement it

Step	Where	What we build
1	Supabase	Table `assistant_usage` (business, time, kind, input / cached / output tokens, model, real cost in $, credits). On `clients`: `credits_balance`, `monthly_allowance`, `allowance_resets_at`. Two safe database functions: `spend_credits` (deducts atomically, never below zero) and `add_credits` (only the payment webhook may call it).
2	n8n engine · AI step	Replace the AI Agent node with one direct call to the Claude Messages API (HTTP Request). This returns exact token counts, lets us switch on **prompt caching**, and lets us put the services list into the prompt so price questions need one call, not two.
3	n8n engine · routing	A small rule picks the model: short, simple questions → Haiku 5.5; buying, quotes, photos, long chats → Sonnet 5.5. Answers capped at a set length; chat memory kept to the last 8 messages.
4	n8n engine · gate	Before the AI: if `credits_balance` is 0, send the friendly no-AI reply and alert the owner. After the AI: call `spend_credits` and write one `assistant_usage` row. At 80% used, one alert per cycle.
5	Stripe	Products for the monthly plans and top-up packs, with the business id in the payment metadata. The webhook (in the quickstark.tech app) calls `add_credits`, records the payment and sets the next reset date.
6	n8n · schedule	Nightly job: reset allowances that are due, total each business's cost and margin, flag outliers to you.
7	Hosting	Move n8n to self-hosting (Railway, next to Evolution) and filter the bot's own outgoing messages out of the Evolution webhook, so each customer message is one run.
8	Dashboard	Usage bar, chart, top-up and auto top-up for owners; cost and margin table for you (part of the dashboard build, Option A).

Rollout

Weeks 1–2 · Measure
Steps 1–2 only: log real tokens and cost on every reply. Nobody is charged yet.

Weeks 3–4 · Optimise
Steps 3 and 7: caching, model routing, self-hosted n8n. Re-run this calculation with real numbers.

Then · Charge
Steps 4–6 and 8: allowances, top-ups and alerts go live with prices set from the measured costs.

All figures on these pages are estimates from a small sample of real replies, today's list prices and an assumed message mix. Prices for Claude, n8n, Railway and the naira rate change; re-check before publishing prices.

9 · RISKS AND DECISIONS

Risks to plan for

Risk	What we do about it
WhatsApp restricting numbers (Evolution links like WhatsApp Web)	Fine for small businesses. Keep message volume natural, no bulk sending. For larger clients, add the **official WhatsApp Business API** later as a premium option; the engine can support both.
n8n execution limits as businesses grow	Self-host n8n before charging for AI (see section 8): a flat monthly fee instead of paying per run. Usage shown in your admin view.
One business seeing another's data	Row-level security on every table, tested on a database copy before going live.
Unpaid bots using AI	Bots answer only while status is *live* and payment is current.
Changes breaking the live bot	The engine is not changed; the dashboard only writes data the engine already reads.

Decisions for you

1. Option A or B? Recommended: A, inside quickstark.tech.

2. When does a bot go live? Recommended: after payment. Alternatives: after your approval, or a short free trial (for example 7 days or 100 messages).

3. Plans and prices in the dashboard Use the current packages (WhatsApp ₦100k, Starter ₦250k, Business ₦400k, Store/Booking quoted, Care plan ₦20k/month) or add monthly bot plans.

4. Your admin view Recommended: a simple admin page listing every business, status, usage and payments, so you never need n8n for day-to-day work.

5. AI usage pricing Recommended: keep the one-time setup fee and add a monthly AI plan (Lite ₦20k, Standard ₦45k, Busy ₦100k) plus top-ups, after 2–4 weeks of measuring real costs.

Next step: review this plan, answer the five decisions, and say “build Option A” (or B). Phase 1 can start the same day.